

Setting Up a Test for a Population Proportion

AP Statistics · Topic 6.4 · Student Packet

OBJECTIVES

Math Objective	Set up a one-proportion z -test by (1) stating H_0 and H_a using correct parameter notation p , (2) verifying all three conditions (Random, 10%, Large Counts), and (3) computing the test statistic $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$.
Language Objective	Students will use the frame “If the true proportion is _____, the probability of getting a sample proportion as extreme as _____ by chance alone is _____.” to interpret a p -value in context.
Essential Understanding	A significance test answers ONE question: is the observed sample proportion far enough from p_0 that we can reject the null? The three-condition check is not busywork — it validates that the z -statistic formula is trustworthy for this sample.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students move from informal “is this surprising?” thinking (Do Now) to the formal four-step AP test structure. Explore runs the full hypothesis-testing sequence on one continuous context so each step builds on the last.
Essential Question	How do we decide whether an observed sample proportion is statistically surprising, and what conditions must hold before we can trust that decision?
Materials	Pencils; student packet; calculator (normalcdf / z -table for optional reinforcement). No Chromebooks required during Explore.

DO NOW

Do Now

A coin is flipped 80 times and lands heads 52 times. (a) What is $\hat{p}$, the sample proportion of heads? (b) If the coin were fair, what would you expect $\hat{p}$ to equal? (c) Does your result from (a) convince you the coin is unfair? Explain in one sentence using symbolic form.

LAUNCH

[BOOK] SETUP RULES: ONE-PROPORTION z -TEST (keep printed on your page)

Four AP Steps (write these before every significance test):

- Hypotheses:** State $H_0: p = p_0$ and $H_a: p \neq p_0$ (or $<$, $>$). Always about the *parameter* p .
- Conditions:** (i) Random sample/assignment. (ii) Independence: $n < 10\%$ of population. (iii) Large Counts: $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

3. **Test Statistic:** $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$. Use p_0 , NOT $\hat{p}$, in the denominator.

4. **Conclusion:** State whether you reject or fail to reject H_0 , in context, at the given α level.

[WARN] COMMON ERRORS (both appear on AP Free Response!)

ERROR 1: Writing $H_0: \hat{p} = p_0$. The null is about the *parameter* p , never the *statistic* $\hat{p}$.

ERROR 2: Using $\hat{p}$ instead of p_0 in the standard error denominator. The formula assumes the null is true.

ERROR 3: Concluding “we proved H_0 ” when you fail to reject. We only ever *fail to reject* — never accept H_0 .

EXPLORE

Bank-item labels (in order):

1. Practice #1 — State the hypotheses
2. Practice #2 — Check all three conditions
3. Practice #3 — Compute the test statistic z
4. Practice #4 [STAR] PERFORMANCE TASK — Design your own 1-prop test

Practice #1 — State the hypotheses

A city council claims that 60% of residents support a new park. A survey of 120 randomly selected residents finds 64 who support the park. State the null and alternative hypotheses (using parameter p) for a two-sided test of the council's claim.

Practice #2 — Check all three conditions

Using the park-survey context from Practice #1 ($n = 120$, $p_0 = 0.60$): verify all three conditions for a one-proportion z -test. Show each check symbolically.

Practice #3 — Compute the test statistic z

Still using the park-survey context ($\hat{p} = 64/120$, $p_0 = 0.60$, $n = 120$): compute the test statistic using the formula $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$. Leave intermediate values in symbolic form until the final step.

Practice #4 [STAR] PERFORMANCE TASK — Design your own 1-prop test

[STAR] PERFORMANCE TASK: Choose a real-world claim about a population proportion (e.g., school attendance rate, recycling rate, seat-belt use). (A) State H_0 and H_a with the correct parameter p and direction. (B) Describe a realistic sample design and check whether all three conditions would plausibly be met. (C) Explain why a researcher would choose a one-sided vs. two-sided alternative for your scenario.

SHARE / SUMMARY

[SEARCH] BRIDGE TO NEXT LESSON

Next class we compute the p -value (using a z -table or `normalcdf`) and write the full four-step conclusion. The setup you practiced today is the foundation. If you are unsure of any step, re-read the Setup Rules callout before next class.

Sentence frames

Direction justification: "I chose a (one/two)-sided alternative because I wanted to detect whether the proportion was specifically than p_0 , not just different."

Conditions check: "The Large Counts condition is met because $np_0 = \quad \geq 10$ and $n(1-p_0) = \quad \geq 10$."

EXIT

The null hypothesis for a one-proportion z -test always states

The condition I found hardest to verify today was because

I distinguish a one-sided from a two-sided alternative by

OPTIONAL REINFORCEMENT (not collected)

Optional #1 — Interpret the p -value in context

The park-survey test yields $z \approx -1.49$ and p -value ≈ 0.136 . Interpret this p -value in context using the sentence frame: “If the true proportion of residents who support the park is ____, the probability of getting a sample proportion as far from ____ as ____ by chance alone is ____.” Then state your conclusion at the $\alpha = 0.05$ level.